



MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. EXAMINATIONS - FEBRUARY/MARCH 2023

SUBJECT: MAT 1171/ MAT_1171 - ENGINEERING MATHEMATICS - I
(MAKEUP)

Answer ALL questions.

ENGINEERING MATHEMATICS - I [MAT 1171-PHY]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

- 1A) From the following table of values of x and y , obtain $\frac{dy}{dx}$ for $x = 1.2$.

x :	1.0	1.2	1.4	1.6	1.8	2.0	2.2
y :	2.7183	3.3201	4.0552	4.9530	6.0496	7.3891	9.0250

(4)

- 1B) Solve $\frac{dy}{dx} - \frac{2y}{1+x} = x^2 + 2x + 1$ (3)

- 1C) Using Lagrange's interpolation, calculate the profit in the year 2000 from the following table:

Year	1997	1999	2001	2002
Profit in Lakhs of ₹	43	65	159	248

(3)

- 2A) Using Runge-Kutta method of order 4, solve $y' = x + y^2$ with $y(0) = 1$ at $x = 0.1, 0.2$ in steps of length $h = 0.1$. (4)

- 2B) Solve $(2xy + y e^x) dx + (x^2 + e^x) dy = 0$ (3)

- 2C) Using Simpson's $\frac{1}{3}$ rule, evaluate $\int_0^1 \frac{x^2}{1+x^3} dx$ with $h = 0.25$. (3)

- 3A) Using Gauss elimination method, solve the following system of linear equations:

$$\begin{aligned} 5x - y + z &= 10 \\ 2x + 4y &= 12 \\ x + y + 5z &= -1 \end{aligned}$$
(4)

- 3B) Solve $(D^2 - 6D + 10)y = \cos 2x$ (3)

- 3C) Using Newton-Raphson method, find the real root of $x^3 - 5x + 3 = 0$ in (1,2). Correct to four decimal places. (3)

- 4A) Using Gram Schmidt Orthogonalization process, construct an orthonormal basis from the set of vectors $\{(1, 1, 1), (2, -1, 2), (1, 2, 3)\}$ for $\mathbb{R}^3$. (5)
- 4B) Using Rayleigh's power method, find the largest eigenvalue and the corresponding eigenvector of the matrix $\begin{bmatrix} 1 & 3 & -1 \\ 3 & 2 & 4 \\ -1 & 4 & 10 \end{bmatrix}$ by taking the initial vector $[1 \ 1 \ 0]^T$. Perform four iterations. Correct to 2 decimal places. (5)
- 5A) Using Taylor series method, find $y(4.1)$ and $y(4.2)$ to four decimal places, given $5x \frac{dy}{dx} + y^2 - 2 = 0$ with $y(4) = 1$. (5)
- 5B) Using Gauss-Jordan method, find the inverse of following matrix

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 2 \\ 3 & 1 & 1 \end{bmatrix}$$
 (5)

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